

A Field Guide to the WA Electricity Market Data

What every dataset in this project actually measures, why it looks the way it does, and the handful of concepts you need before you trust a single number out of it.

6

DATASETS COVERED

3

YEARS OF HISTORY

0

PRIOR KNOWLEDGE ASSUMED

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Where this data comes from

Every file in this project comes from the **Australian Energy Market Operator (AEMO)**, the organisation that runs Australia's electricity markets. AEMO doesn't own power stations or wires — it's the referee. Every five minutes it looks at how much electricity people want, tells generators how much to produce, and publishes the price that resulted. All of that gets logged and published as open data, which is what this project reads.

DEFINITION

WEM (Wholesale Electricity Market) is the market AEMO runs specifically for Western Australia. **SWIS (South West Interconnected System)** is the physical grid it operates on — the network of poles, wires, and power stations connecting Perth, the South West, and the Wheatbelt.

The important thing that makes WA data simpler to reason about than most other regions:

WA's grid is an electrical island

Unlike the eastern states — where New South Wales, Victoria, Queensland, and South Australia are all wired together into the National Electricity Market (NEM) and can ship power to each other when one state runs short — the SWIS has **no transmission link to anywhere else**. It is thousands of kilometres from the nearest other grid, across mostly empty desert. Whatever electricity WA uses, WA (and only WA) has to generate at that exact moment. There is no "import power from a neighbour" escape valve. This is why battery storage and demand-side flexibility matter so much in this data: there's nowhere else for the balancing act to happen.

WA doesn't observe daylight saving

Western Australia stopped observing daylight saving in 2009. That's a small political fact with a big practical consequence for this project: every timestamp in every file is a clean, fixed **UTC+8 (Australian Western Standard Time, AWST)** offset, all year round. There's no "the clocks jumped forward an hour" gap in spring and no duplicated hour in autumn — a problem that trips up almost every other Australian or US dataset. One less thing to debug.

WHY IT MATTERS FOR THIS PROJECT

Because the grid is isolated, demand and supply have to balance locally, in real time, using only what's connected to the SWIS: gas and coal generators, wind and solar farms, rooftop solar on houses, and — increasingly — grid-scale batteries. That's exactly the cast of characters the six datasets in this guide describe.

Two units you must never confuse

Almost every mistake a beginner makes with this data traces back to mixing up two units that sound similar but measure completely different things: **megawatts (MW)** and **megawatt-hours (MWh)**.

DEFINITION

Power (MW) is a *rate* — how fast energy is flowing right now.

Energy (MWh) is an *amount* — how much energy flowed in total over some period of time.

Think of a car. Its speedometer reads in km/h — a rate, telling you how fast you're going at this instant. Its odometer reads in km — a total, telling you how far you've actually travelled. Power (MW) is the speedometer. Energy (MWh) is the odometer. You cannot read one off the other without also knowing how long you travelled for.

```
Energy (MWh) = Power (MW) × Time (hours) // e.g. a 100 MW generator running flat out for
30 minutes (0.5 h) produces 50 MWh
```

WHERE THIS BITES IN THE ACTUAL DATA

The battery SCADA files (Section 4.4) report `Average MWh` — energy delivered over a 5-minute dispatch interval, *not* a power reading. To compare it with every other series in this project, which are all in MW, you have to convert: since 5 minutes is 1/12 of an hour, multiply the MWh figure by 12 to get average MW. The project's loader (`load_facility_scada`) does this for you automatically and hands back an explicit `mw` column — but it's worth knowing why that conversion exists, because plenty of public AEMO tables don't do it for you.

Two clocks running at once

This project mixes data recorded on two different clocks, and understanding the difference matters for anything you aggregate or compare.

CLOCK	LENGTH	USED BY	WHAT "ONE ROW" MEANS
Dispatch interval	5 minutes	Demand & withdrawal, distributed PV, battery SCADA	AEMO's physical control loop — the grid is actually rebalanced this often
Trading interval	30 minutes	Reference trading price	The financial settlement window — the price generators actually get paid

A trading interval is simply six dispatch intervals bundled together. This project's helper function `to_trading_intervals()` exists specifically to average the 5-minute series up to 30 minutes so they can be compared against price on equal footing.

LABELLING CONVENTION

AEMO labels a trading interval by its *start* time. The row stamped 12:00 covers everything from 12:00 up to (but not including) 12:30 — i.e. the six dispatch intervals at 12:00, 12:05, 12:10, 12:15, 12:20 and 12:25. It is easy to assume a timestamp marks the *end* of a period instead; AEMO does the opposite.

All timestamps are local Perth time (AWST)

Every timestamp column in every file — demand, price, PV, SCADA, temperature — is already in AWST (UTC+8), and (per Section 1) that offset never changes across the year. You do not need to do any timezone conversion to compare series against each other. You do, however, need to know that the underlying *files* are cut on UTC day/year boundaries rather than local ones — see Section 6.

SECTION 04

The six datasets, one by one

Each of the following is one AEMO data product this project loads and joins together. For each one: what it physically measures, the concept you need to interpret it correctly, and the single most common way to misread it.

4.1 Operational Demand & Withdrawal

OperationalDemandWithdrawal-{2023..2026}.csv

Resolution: **5 min**

Unit: **MW**

Rows loaded: **307,008**

This is the closest thing to "how much electricity is Western Australia using right now." But the file actually contains *three* related demand numbers, not one, and the difference between them is one of the most useful facts in the whole project.

THE THREE COLUMNS, DEFINED

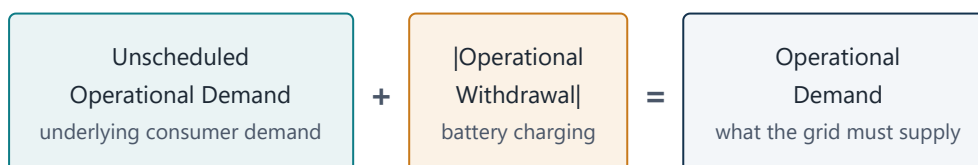
Operational Demand — the total power that grid-connected generators are sending out to meet, including transmission and distribution losses along the way. Crucially, this is *net of rooftop solar*: if your house makes its own solar power and uses it, that demand never touches the grid and never shows up here.

Unscheduled Operational Demand — Operational Demand with one more thing subtracted: the power being drawn by battery storage while it charges. This is the closest number to genuine, underlying *consumer* demand — homes, factories, offices, shops.

Operational Withdrawal — the power batteries are drawing from the grid to charge. Always zero or negative in this file (range: -1,158 to -0.40 MW), because "withdrawal" is measured as a subtraction from supply.

These three aren't independent — this project's data-quality audit verified the relationship holds exactly (to rounding error) across all 307,008 five-minute intervals in the dataset:

```
Operational Demand = Unscheduled Operational Demand - Operational Withdrawal //  
withdrawal is ≤ 0, so this is really: Operational Demand = Unscheduled Demand + battery  
charging
```



WORKED EXAMPLE

Suppose at 2pm, Unscheduled Operational Demand is 2,000 MW (homes and businesses drawing power) and batteries across the state are charging at a combined 150 MW. Operational Withdrawal is recorded as -150 MW. Operational Demand — the figure the grid actually has to generate — is $2,000 - (-150) = \mathbf{2,150 \text{ MW}}$. The batteries added 150 MW of extra load on top of what people were actually using.

WHY IT MATTERS

Because they now move in *opposite* directions at times, "minimum demand" — a headline grid-planning statistic — means something different depending on which of the two demand series you quote.

Operational Demand is flattered upward by battery charging; Unscheduled Operational Demand strips that back out to show what consumers alone are doing. A responsible analysis reports both.

4.2 Distributed PV (Rooftop Solar)

distributed-pv-new-{2023..2026}.csv

Resolution: **5 min**

Unit: **MW**

Rows loaded: **307,007**

Not every solar panel in WA is a big, purpose-built solar farm that AEMO can watch directly. Hundreds of thousands of houses have small rooftop panel systems that generate power invisibly, from the grid operator's point of view — the power is made and consumed (or exported) locally, behind each household's electricity meter.

DEFINITION

Behind-the-meter generation is electricity produced on the customer's side of their meter — rooftop solar being the classic example — as opposed to **grid-scale** generation from a registered power station that AEMO dispatches directly. Because AEMO cannot see behind-the-meter generation directly, this file is an **estimate**, modelled from things like weather, known installed rooftop capacity, and time of day/year — not a metered reading.

This is the missing piece that connects to Section 4.1: Operational Demand is *net* of rooftop solar, meaning rooftop generation has already been subtracted out before that file was even published. Distributed PV is how you get it back, if you want to reconstruct *gross* underlying consumption — what people would be drawing from the grid if nobody had solar panels at all.

COMMON PITFALL

It's tempting to treat Distributed PV like any other generator's output. But because it's estimated rather than metered, and because it's the reason WA's demand curve dips so dramatically around midday (the so-called "duck curve" effect, where grid demand falls as home solar peaks and then surges again in the evening), small errors in this estimate can make demand look more volatile than it really is. Treat it as directionally reliable, not to-the-megawatt precise.

4.3 Reference Trading Price

ReferenceTradingPrice-{2023..2026}.csv

Resolution: **30 min**

Unit: **\$/MWh**

Rows loaded: **51,168**

This is the wholesale price of electricity in WA — what it cost, on average, to buy one megawatt-hour of power during each half-hour trading interval. It's not what households pay on their power bill (that's a separate, much smoother retail tariff); it's the raw market-clearing price that gets paid to generators and, ultimately, flows through to retail prices over time.

WHY PRICES CAN GO NEGATIVE

A price below \$0/MWh means generators are effectively *paying* someone to take their electricity. This isn't a data error — it happens in real markets when there's more low-cost generation (typically wind and solar) available than demand needs, and some generators would rather pay a small fee than shut down and restart later. It's a genuine, if unusual, market outcome.

Administered price limits change over time

AEMO sets an official floor and ceiling that prices are not allowed to cross, and — importantly for comparing years — **those limits have moved**:

YEAR	MIN PRICE	TIMES HIT	MAX PRICE	TIMES HIT
2023 (Oct–Dec)	−\$716.71	2	\$738.00	43
2024	−\$1,000.00	44	\$738.00	123
2025	−\$150.52	1	\$1,100.00	1
2026 (to Aug)	−\$16.72	1	\$1,000.00	1

READING THIS TABLE CORRECTLY

In 2023–2024, the maximum price landed on *exactly* \$738.00 dozens of times — a sign the price was hitting a hard administered ceiling, not that the market happened to produce that number repeatedly. From 2025 onward the extreme prices are unique, one-off values well under the new higher ceiling — the market simply never got that stressed again. Comparing "price volatility" across years without knowing this makes 2024 look artificially wilder than 2026 on the downside, and masks how rare extreme prices became later.

Not every price is "real"

The file also carries a `Failure Reason` column. Most rows are blank (the price was set normally), but some intervals are flagged — the most important flag being `MarketAnalystOverride`, meaning a human manually set the price rather than letting the automated market engine calculate it (used on roughly 1.3–1.8% of intervals in 2025 and 2026). Any serious price analysis should be checked with these overridden intervals excluded.

4.4 Facility SCADA (Battery Storage)

FacilityScada-YYYY-MM.csv (filtered to battery facilities)

Resolution: **5 min**

Unit: **MWh** → **MW**

Rows loaded: **1,096,453**

DEFINITION

SCADA stands for *Supervisory Control and Data Acquisition* — the industrial monitoring systems that watch physical equipment (here, grid-scale batteries) in real time and report back what it's doing. This file is the closest thing in the project to ground-truth sensor data, one row per battery facility per 5-minute interval.

Two things make this dataset different from the others: the unit conversion covered in Section 2, and a sign convention that tells you which direction the battery is moving power.

SIGN	MEANING
Positive MW	Discharging — the battery is sending power out to the grid
Negative MW	Charging — the battery is drawing power from the grid to store

A quirk worth knowing: "System Size (MW)" isn't the battery's power rating

You might expect a battery registered at, say, 300 MW to be able to charge at up to 300 MW and discharge at up to 300 MW. It's actually recorded as the *sum* of both directions:

FACILITY	MAX DISCHARGE	MAX CHARGE	OBSERVED SPAN	REGISTERED "SYSTEM SIZE"
KWINANA_ESR1	100.7 MW	−100.1 MW	200.8 MW	200 MW
COLLIE_ESR1	200.0 MW	−200.1 MW	400.1 MW	400 MW
COLLIE_BEES2	300.1 MW	−300.0 MW	600.0 MW	600 MW

WHY THIS MATTERS

Quoting `System Size (MW)` as "how big is this battery" overstates its real power rating by roughly 2×. `COLLIE_BEES2` is a **300 MW** battery (it can charge at up to 300 MW *and* separately discharge at up to 300 MW) — not a 600 MW one. The registered figure is a two-sided span, not a one-way rating. This project's registry loader keeps the raw figure but this table should be consulted before quoting battery capacity anywhere.

4.5 Facility Registry

facilities.csv

Resolution: **snapshot**

Rows loaded: **176**

Every other dataset in this project refers to power stations, batteries, and large loads by a short **facility code** (e.g. COLLIE_BESS2, ALCOA_KW_IL). This file is the lookup table that turns those codes into something meaningful: who owns the facility, what class of participant it is, and (nominally) how big it is.

THE ONE THING THIS FILE IS MISSING

There is no "technology" or "fuel type" column — nothing that directly says "this one is wind" or "this one is a battery." This project infers it instead from a naming convention in the facility code's suffix: _WF for wind farm, _PV/_SF for solar, _ESR/_BESS for storage, _GT/_U#/_G# for thermal (gas/coal). This is a **stated assumption**, not something AEMO officially certifies — and it only successfully classifies 61 of the 176 registered facilities. The rest are mostly Demand Side Programme registrations (large industrial loads that get paid to reduce consumption on request) whose codes don't follow this pattern. It is reliable, however, for the seven storage facilities this project's battery analysis actually depends on.

4.6 Facility Maximum Daily Temperature

Resolution: **daily**

Unit: **°C**

facility-temperature-{2024..2026}.csv

This is the odd one out: everything else in the project is about electricity, but this file is weather — the maximum temperature recorded near each facility on each day. It exists because temperature is one of the strongest drivers of electricity demand that has nothing to do with the market itself.

WHY IT'S HERE

Hot days push air-conditioning load up sharply, which is exactly the kind of demand spike that can push wholesale prices toward their ceiling (Section 4.3) and lean hardest on battery storage (Section 4.4) to smooth things out. Joining this file against demand and price lets an analysis ask "is this price spike a heatwave story or something else?" instead of treating every spike as unexplained.

Resolution here is daily, not 5-minute or 30-minute — so this is a variable you join *onto* the finer-grained series (typically by matching date and, loosely, region), not one you analyse at the same granularity as the rest.

How the six datasets connect

None of these files is interesting entirely on its own — the project's actual insights come from lining them up against each other. Here's the mental map:

DATASET	ROLE IN THE STORY
Distributed PV	Rooftop solar that quietly reduces what the grid has to supply, before anyone else in the chain even sees it.
Operational Demand & Withdrawal	What's left for the grid to generate, split apart into "genuine consumer use" vs. "batteries topping themselves up."
Facility Temperature	The weather variable that explains a large share of <i>why</i> demand moves the way it does, day to day.
Facility SCADA (batteries)	The moment-by-moment record of storage charging (adds to demand) and discharging (acts like extra generation), matched against the registry to know which facility is which.
Facility Registry	The reference table that turns every facility code appearing elsewhere into an owner, a class, and an (inferred) technology.
Reference Trading Price	The financial outcome of all of the above: what it cost to keep supply and demand balanced in each half-hour.

A typical question this project can answer sits at the intersection of several of these: "*On the hottest days, how much did battery charging add to Operational Demand, and did that push the trading price toward its ceiling?*" — which touches temperature, demand/withdrawal, SCADA, and price all at once.

Reading the raw files without getting burned

These are file-format quirks, not conceptual ones — but they've each caused real bugs in this project's history, so they're worth knowing before you open a CSV directly instead of going through the loaders in `src/wa_data.py`.

1. Files are cut on UTC boundaries, but stamped in local time

`OperationalDemandWithdrawal-2025.csv` doesn't span the 2025 calendar year — it runs from 2025-01-01 08:00 to 2026-01-01 07:55 *local* time. The file is actually split at UTC midnight, which is 8am in Perth. Building a "2025 total" from that one file alone silently drops the first eight hours of January and picks up eight hours of the following January. Always concatenate all available files first, then slice by local calendar date — never trust one file to equal one year.

2. Rows arrive newest-first

The interval files list the most recent timestamp in row 1, oldest last — the opposite of what most people expect from a time series. The project's loaders re-sort everything into ascending order.

3. Dates are written day first

Interval files use `dd/MM/yyyy` (e.g. `01/09/2026` means 1 September, not 9 January) — the Australian convention, not the US `MM/dd/yyyy` one. `facilities.csv` and the temperature files use plain ISO dates (`2026-09-01`) instead. It's the same project, two different date formats depending on which file you're in.

4. "Extracted At" is not a revision log

Every file has an `Extracted At` column, but it's only ever filled in on the *first* row of the file — it's a single watermark saying "this whole file was downloaded at this moment," not a per-row record of when that particular reading was last revised or corrected.

5. Downloads can silently truncate

When many monthly battery SCADA files are downloaded in parallel, a network hiccup can cut a file off mid-line. If the cut happens to land exactly on a line boundary, the file looks like valid (just short) data rather than obviously broken — nothing will throw an error. This project's fetch script checks each file's final line against the expected pattern and re-downloads automatically if it looks truncated.

Glossary

Every term used in this guide, gathered in one place for quick lookup.

AEMO

Australian Energy Market Operator — the body that runs the electricity market and publishes this data.

AWST

Australian Western Standard Time, UTC+8. WA's only timezone — it never changes for daylight saving.

Behind-the-meter

Electricity generated and used on the customer's side of their meter (e.g. rooftop solar) — invisible to the grid operator's direct measurements.

Dispatch interval

The 5-minute cycle AEMO uses to physically balance the grid — the finest time resolution in this project.

Failure Reason

A flag on the price file noting when a trading interval's price wasn't set by the normal automated process (e.g. a manual override).

Grid-scale

A generator or battery large enough, and formally registered, to be directly dispatched and monitored by AEMO — as opposed to small, invisible rooftop systems.

MW (megawatt)

A unit of power — a rate of energy flow, measured at an instant. The "speedometer" unit.

MWh (megawatt-hour)

A unit of energy — a total amount, accumulated over time. The "odometer" unit.

NEM

National Electricity Market — the interconnected grid and market covering the eastern Australian states. Not connected to WA's grid at all.

Operational Demand

Total sent-out power from grid-connected generation, net of rooftop solar, including network losses.

Operational Withdrawal

Power drawn from the grid by scheduled loads (in practice, battery charging). Always ≤ 0 in this project's files.

SCADA

Supervisory Control and Data Acquisition — real-time sensor/monitoring data from physical equipment, here grid-scale batteries.

Spot price / Reference Trading Price

The wholesale market-clearing price of electricity, published every 30-minute trading interval, in \$/MWh.

SWIS

South West Interconnected System — the physical electricity grid covering WA's populated South West, operated as an electrical island with no link to any other grid.

Trading interval

The 30-minute window over which electricity price is settled — six dispatch intervals bundled together, labelled by its start time.

Unscheduled Operational Demand

Operational Demand with battery charging subtracted back out — the closest available estimate of genuine, underlying consumer demand.

WEM

Wholesale Electricity Market — the market AEMO runs for Western Australia, distinct from the NEM.

Sources: AEMO WA public market data (data.wa.aemo.com.au); WEM Registration Technical Guide v4.2; 2024 WEM Electricity Statement of Opportunities. All figures above are reproduced from this project's own data audit (`scripts/audit.py`), run 1 September 2026 against files downloaded the same day — see `docs/DATA_NOTES.md` for the full technical record.